

Akshat Gupta

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EDUCATION

- **Indian Institute of Technology, Mandi** Himachal Pradesh, India
Bachelor of Technology in Civil Engineering *Aug 2023 – Present*
 - **Technical Coursework:** Data Structures & Algorithms, Machine Learning, Probability & Statistics, Python for Data Science, Statistical Learning, Calculus, Linear Algebra, ODE & Integral Transform

EXPERIENCE

- **EFKON India Pvt. Ltd. (STRABAG)** Gurgaon, India
Research and Development Intern *Dec 2025 – Jan 2026*
 - **Point Cloud Preprocessing:** Engineered a comprehensive data handling pipeline featuring statistical outlier filtering, voxel downsampling, spatial cropping for lane isolation, and RANSAC-based ground plane segmentation.
 - **Data Representation:** Transformed sparse, unstructured 360-degree LiDAR sweeps into structured 2D range images and 3D Bird's-Eye View (BEV) maps to improve segmentation reliability and enable real-time inference on edge devices.
 - **Sensor Integration & Analytics:** Conducted rigorous sensor testing across indoor and toll-plaza environments; utilized Open3D and ROS to validate data quality, identify noise patterns, and evaluate range limitations.
 - **Commercial Deployment:** Delivered a validated, modular pipeline and structured dataset (logged in .pcd format) that accelerated downstream machine learning development for Intelligent Transportation Systems (ITS).

TECHNICAL SKILLS

- **Languages:** C++, Python, Rust, HTML, CSS, JavaScript
- **Technologies/Frameworks:** React.js, Tailwind CSS, Node.js, Express.js, PyTorch, NumPy, Pandas, OpenCV, TensorFlow, Keras, Open3D, ROS
- **Developer Tools:** Git, GitHub, VS Code, Vercel, Firebase, Jupyter Notebook
- **Core Tools & Domains:** Machine Learning, LiDAR Perception, Data Analytics, AutoCAD, SolidWorks

PROJECTS

- **ChronoCapsule** [Live Demo](#) | [Github](#)
React.js, Firebase, Cloudinary, Vercel *July 2025*
 - **Full-Stack Development:** Developed a scalable web platform allowing users to securely architect and schedule the future delivery of digital time capsules containing media and text.
 - **Cloud Integration:** Implemented robust user authentication and a dynamic frontend integrated with Firebase for database operations and Cloudinary for media storage.
- **Add-on for Pilot Hand Gloves** *Aug 2024 – Dec 2024*
Material Science, Conductive Ink
 - **Hardware Innovation:** Engineered a cost-effective, tear-resistant add-on utilizing conductive ink, enabling defense pilots to interact with touchscreens without removing protective gear.
- **Movie Recommender System** [Github](#)
Streamlit, Pandas, Scikit-learn, TMDB API *July 2024*
 - **Machine Learning:** Engineered a content-based recommendation engine leveraging vectorization techniques and cosine similarity algorithms on datasets extracted via the TMDB API.
 - **User Interface:** Designed an interactive web UI with Streamlit to parse data and display dynamic, personalized recommendations.

ACHIEVEMENTS

- Secured 1st Runner Up in the open house project demonstration of Design Practicum (IC202P).
- Awarded a Rs 2.5L grant for an innovative project proposal in Design Practicum [\[Approval\]](#).
- Invited to BRD Chandigarh to explore and contribute to technical hardware projects for the Indian Air Force.